

MONTROSE, CO, USA

WMO: 724765

Lat: 38.506N

Lon: 107.899W

Elev: 1744

StdP: 82.06

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 93013

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-16.1	-12.9	-19.7	0.8	-11.4	-17.5	1.0	-7.3	10.5	4.9	8.4	3.8	3.1	150	0.568

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.5	34.1	15.2	32.9	14.9	31.9	14.6	17.8	27.8	17.0	26.9	16.5	26.4	4.2	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.3	13.5	19.3	14.1	12.5	19.3	13.0	11.6	20.1	56.7	27.0	54.4	26.8	52.4	26.3	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	8.6	7.3	DB	-19.7	36.8	3.0	2.0	-21.8	38.3	-23.5	39.5	-25.2	40.6	-27.4	42.1	(4)
(5)			WB	-20.1	19.1	2.9	1.7	-22.1	20.3	-23.8	21.2	-25.5	22.1	-27.6	23.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	-2.3	0.9	5.8	9.5	14.6	20.6	23.5	21.9	17.4	10.3	3.7	-2.4	(6)	
	DBStd	9.76	4.40	4.46	4.28	4.09	4.21	3.46	2.22	1.96	3.54	4.17	4.52	4.76	(7)	
	HDD10.0	1469	381	254	140	58	11	0	0	0	2	46	193	383	(8)	
	HDD18.3	3312	639	488	388	265	126	19	1	2	55	248	441	641	(9)	
	CDD10.0	1597	0	0	11	44	154	317	419	368	224	57	3	0	(10)	
	CDD18.3	398	0	0	0	0	11	86	162	112	27	1	0	0	(11)	
	CDH23.3	5599	0	0	1	19	312	1449	1985	1311	489	33	0	0	(12)	
(13)	CDH26.7	2402	0	0	0	1	72	662	986	543	135	2	0	0	(13)	
(14)	Wind	WSAvg	3.1	2.4	2.7	3.4	3.8	3.8	3.3	3.1	3.2	3.0	2.7	2.3	(14)	
(15) Precipitation	PrecAvg	248	13	11	18	21	22	15	24	30	31	28	20	16	(15)	
	PrecMax	381	50	25	57	72	63	57	76	82	85	66	70	50	(16)	
	PrecMin	142	1	0	0	0	0	0	1	4	0	0	0	0	(17)	
	PrecStd	55	10	7	15	15	16	15	18	18	23	18	13	10	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.3	16.0	22.5	25.6	32.0	35.6	36.3	34.1	32.1	26.4	19.3	12.9	(19)	
		MCWB	4.0	5.8	8.3	9.8	13.6	15.1	15.9	15.5	15.2	11.8	7.6	4.5	(20)	
	2%	DB	8.7	13.0	19.8	23.0	28.5	33.8	34.8	32.8	29.8	23.9	17.1	9.9	(21)	
		MCWB	2.6	4.4	7.1	8.4	11.9	14.3	15.6	15.2	13.8	10.7	6.8	3.7	(22)	
	5%	DB	7.0	10.9	17.5	21.3	26.9	32.4	33.5	31.6	28.0	21.9	14.8	7.4	(23)	
		MCWB	1.8	3.5	6.0	7.8	11.0	13.5	15.4	15.2	13.4	9.9	5.9	2.1	(24)	
	10%	DB	4.7	8.4	15.1	19.1	24.9	30.9	32.2	30.0	26.5	19.2	12.3	5.0	(25)	
		MCWB	0.6	2.5	5.1	7.2	10.3	13.1	15.3	15.2	12.9	8.9	4.8	0.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.0	6.7	9.0	10.5	14.2	16.8	18.8	19.1	16.8	13.5	8.7	5.7	(27)	
		MCDB	9.4	13.6	20.2	22.3	28.8	30.8	30.0	29.2	25.9	21.2	17.4	11.4	(28)	
	2%	WB	3.4	5.1	7.6	9.4	12.9	15.5	17.9	17.8	15.7	11.9	7.4	4.1	(29)	
		MCDB	7.2	11.9	18.4	20.9	25.4	29.4	28.5	26.9	24.6	20.6	15.4	8.6	(30)	
	5%	WB	2.2	4.0	6.5	8.5	11.9	14.8	17.2	16.9	14.9	10.8	6.3	2.6	(31)	
		MCDB	6.0	9.7	16.2	19.4	23.9	28.6	27.6	26.1	23.5	19.2	13.8	6.9	(32)	
	10%	WB	1.0	2.9	5.5	7.7	10.9	14.0	16.7	16.4	14.1	9.9	5.2	1.2	(33)	
		MCDB	4.4	7.9	14.2	18.0	22.8	27.6	27.0	25.7	23.1	18.0	11.6	4.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.7	12.1	14.4	14.6	15.6	17.6	16.5	15.9	15.9	15.2	13.6	11.5	(35)	
		MCDBR	13.4	15.6	18.2	18.5	18.7	19.6	18.8	18.5	18.5	18.6	17.0	13.3	(36)	
		MCWBR	8.4	8.5	8.7	8.2	7.5	7.4	6.7	6.8	7.2	8.4	9.1	8.2	(37)	
	5% WB	MCDBR	11.5	13.6	16.6	16.8	17.4	17.8	15.2	14.6	14.8	15.8	15.8	12.1	(38)	
		MCWBR	7.3	7.8	8.1	7.6	7.1	6.8	5.3	5.5	6.2	7.5	8.6	7.6	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.243	0.247	0.279	0.303	0.316	0.314	0.361	0.356	0.309	0.281	0.252	0.243	(40)	
	taud		2.579	2.586	2.507	2.469	2.485	2.484	2.406	2.410	2.547	2.593	2.612	2.571	(41)	
	Ebn at Noon		967	1003	990	976	962	959	912	909	944	947	947	941	(42)	
	Edh at Noon		70	80	97	107	108	108	116	113	92	79	68	66	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.63	3.43	4.84	5.95	6.80	7.85	6.93	6.09	5.38	4.04	2.90	2.25	(44)	
	RadStd		0.17	0.25	0.38	0.36	0.53	0.49	0.40	0.28	0.29	0.33	0.21	0.18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	-0.46	N/A	N/A	N/A	N/A	N/A
(47)	Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page